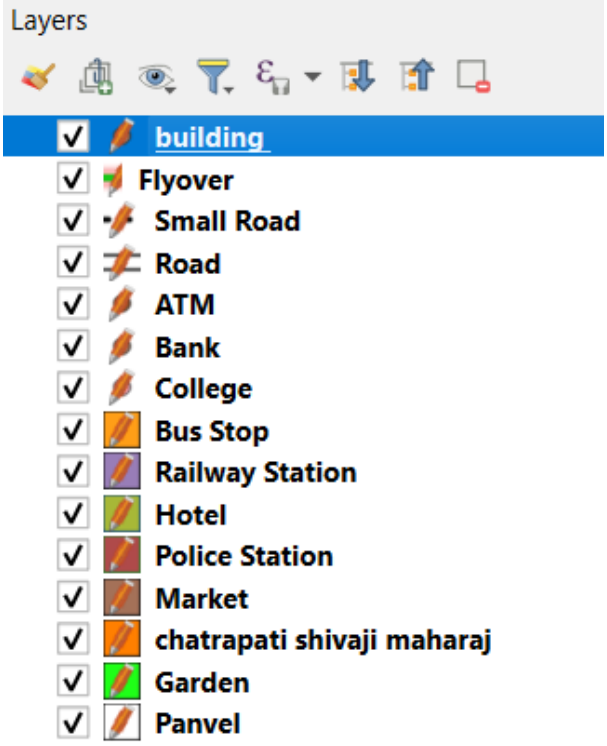
Practical 2

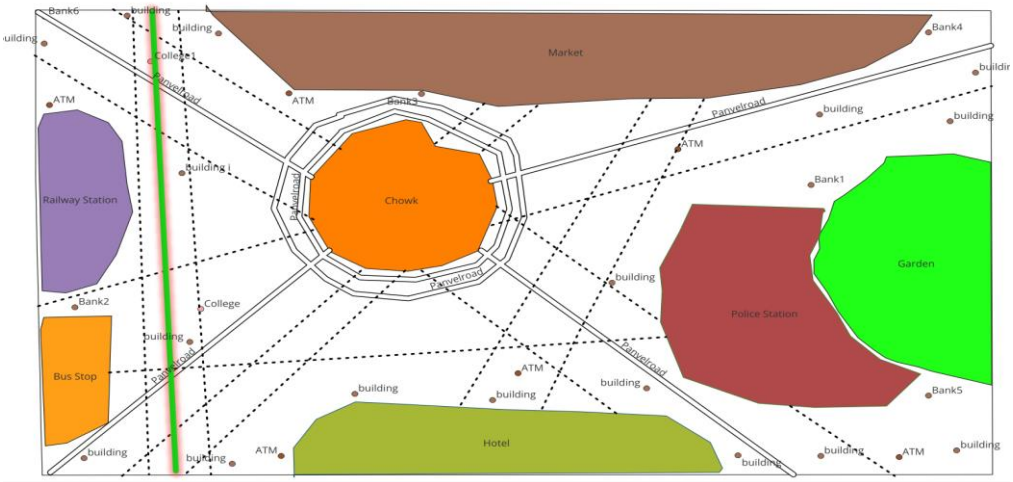
Creating and Managing Vector Data: Adding vector layers, setting properties, formatting, calculating line lengths and statistics

a. Adding vector layers(Polygon, Line, Points)
   - Polygon layers(We have taken 2 layers Matunga, Garden)
   - Line layers(We have taken 3 layers Small_Roads, Road, Flyover)
   - Point layers(We have taken 4 layers Bank,college,restaurants,ATM)

b. Setting properties(Labelling, Symbolism)

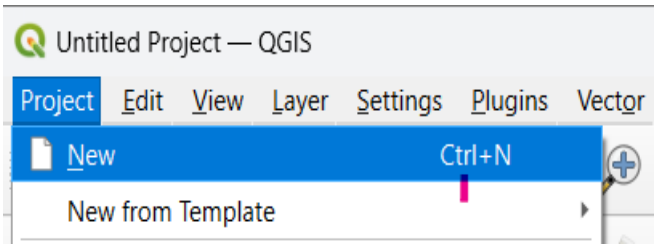


- Our aim is to create map representing a location and its surrounding as follows:

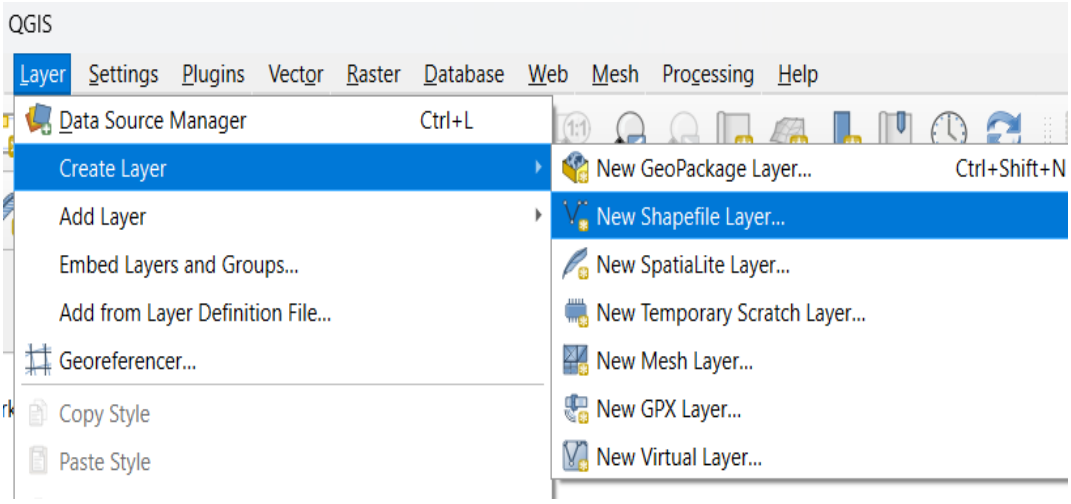


a) Creating Polygon vector layer

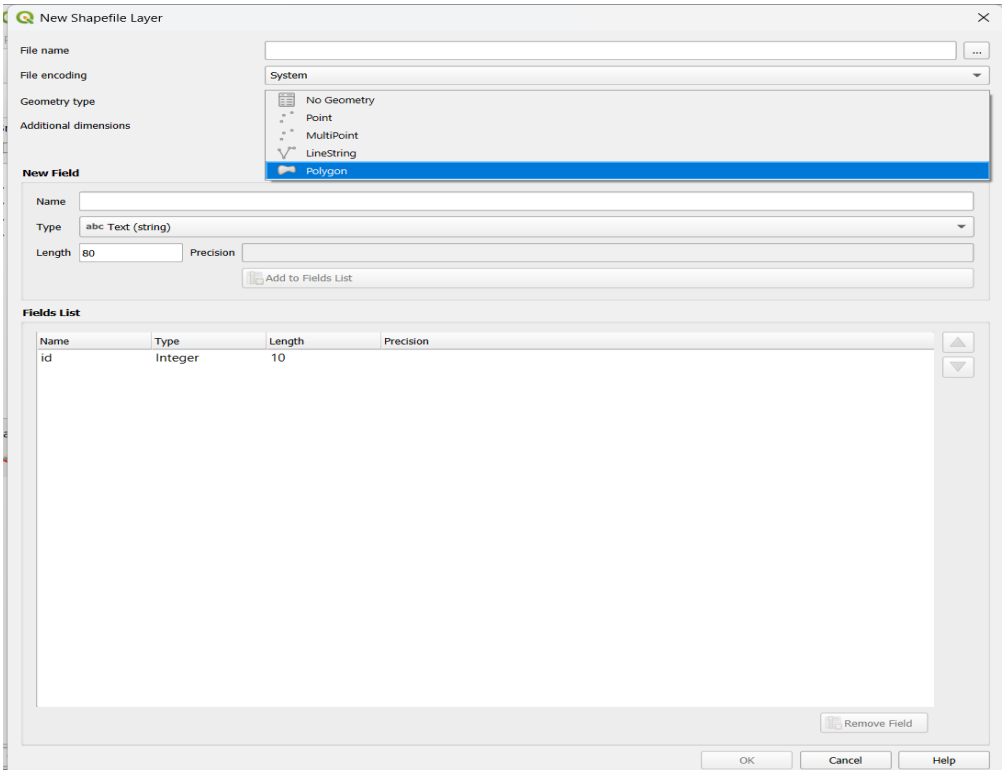
- Select **Project -> New**



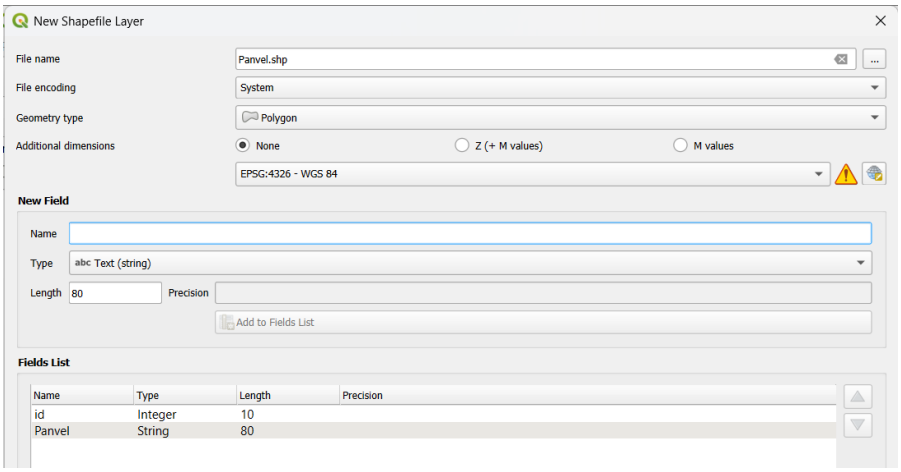
- Select **Layer-> Create Layer-> New Shapefile Layer**

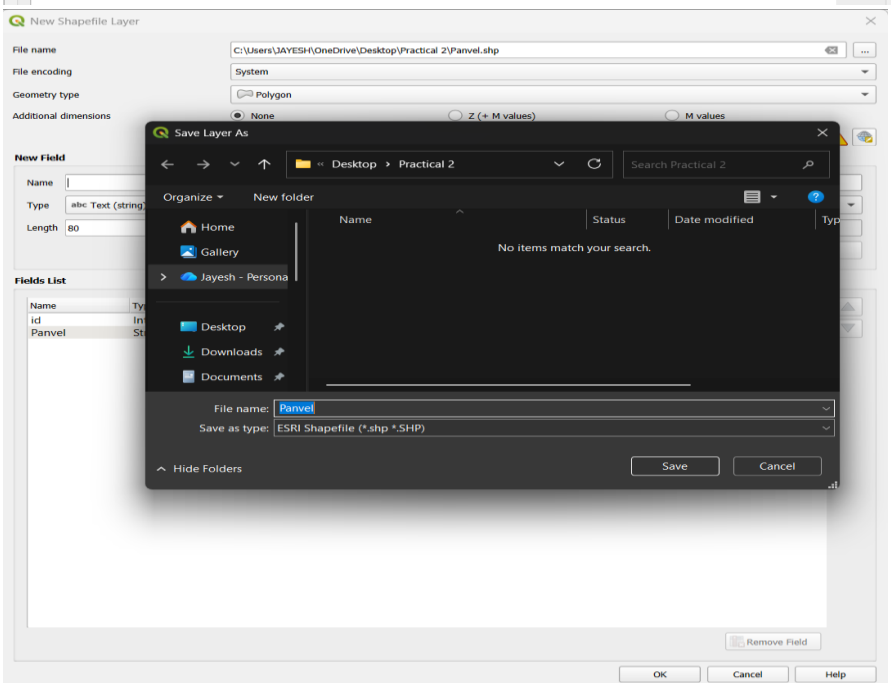


- Following dialog box will appear on the screen. Select polygon option from Geometry type.

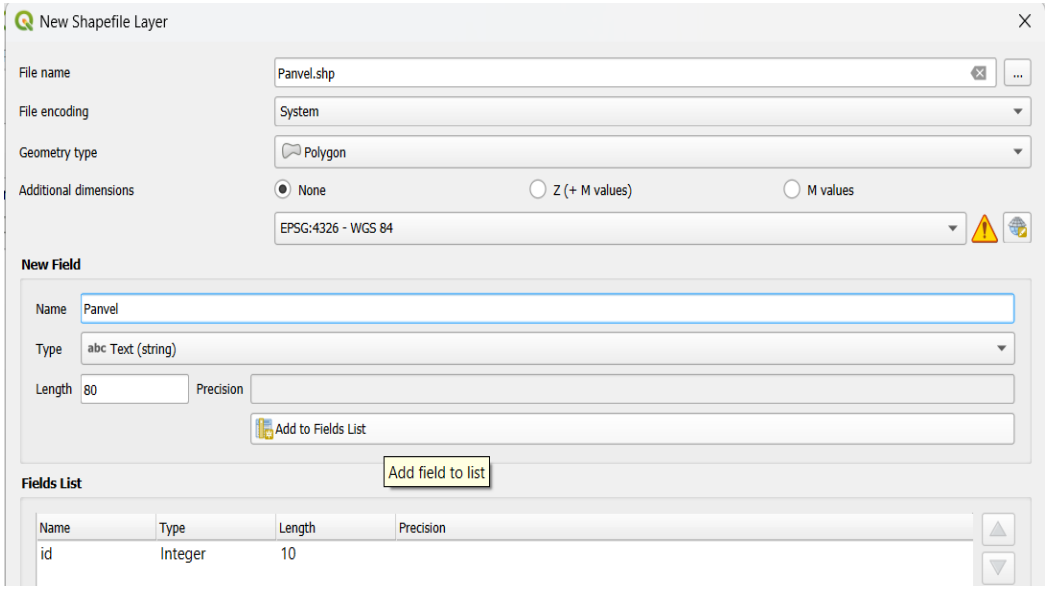


- Fill the appropriate information in each text box.
    - File name:
        1. By default the file will be saved in bin folder.
        2. To avoid it click on following button to change the location of file.

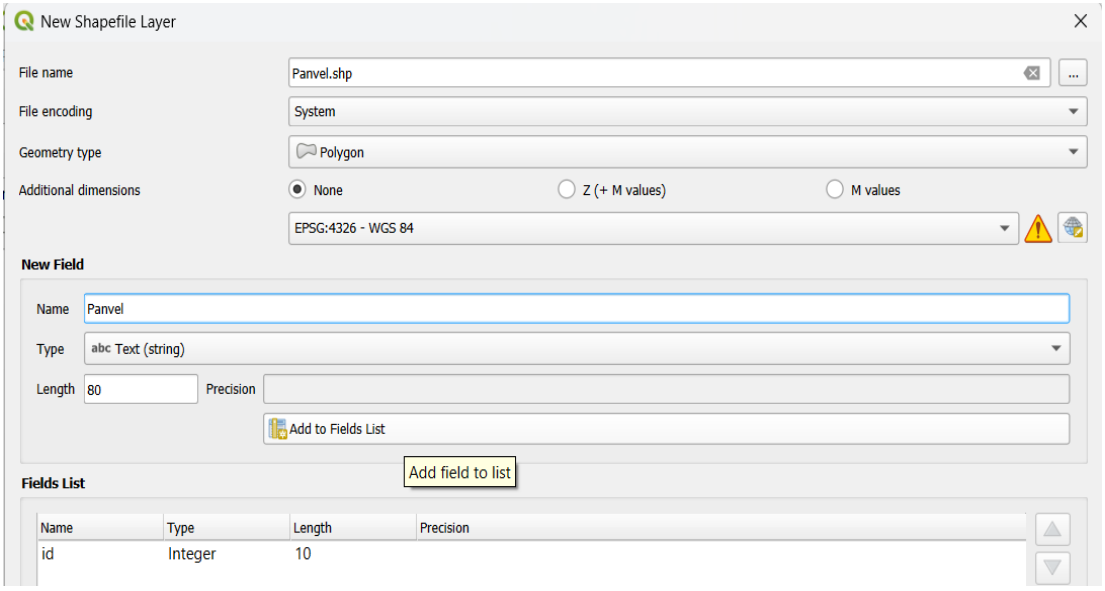




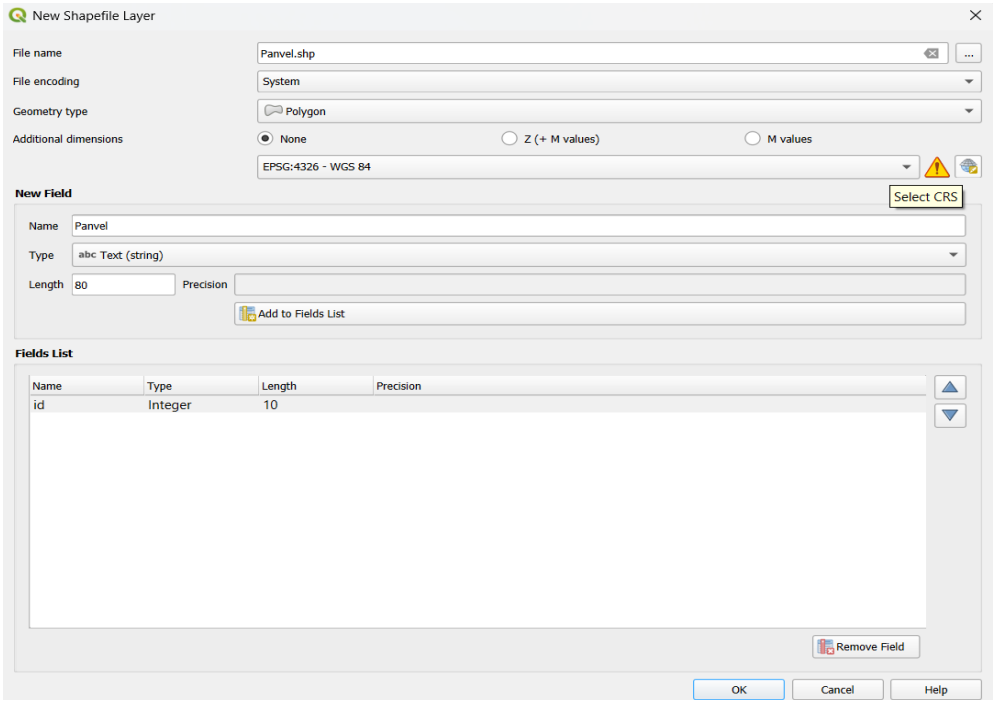
- **Field Panel**
  a. Add the **Attribute** you want to show.(**Column Name** for Table)
  b. Specify **Type(Data Type:** Text Data/Decimal Data/Whole Number/Date) of attribute
  c. Specify the Length of the Attribute. Specify **Precision**(**If Data** Type is **Decimal**)



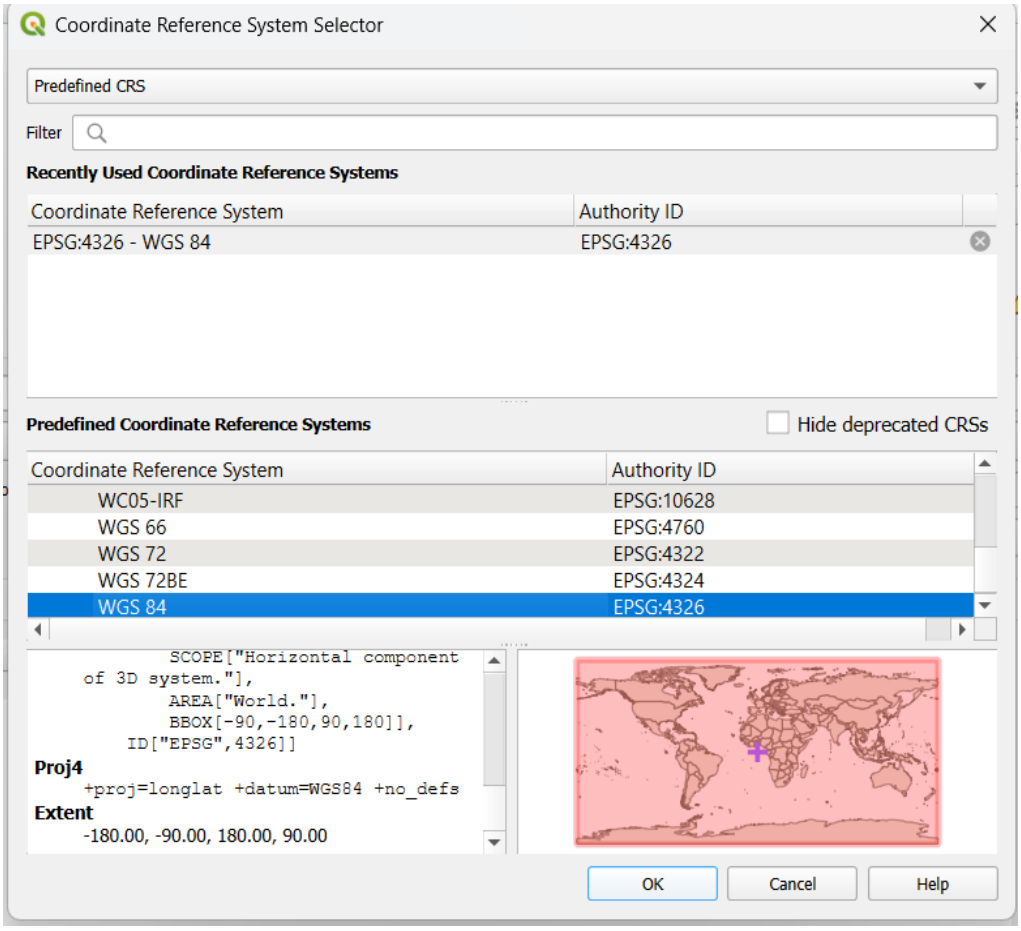
- Click On **Add to Field List** Button.
- You can add as many **Fields** (**Column Name**) as you want for the layer.



- Select Geometry Type as follows:
  - Click on the following button

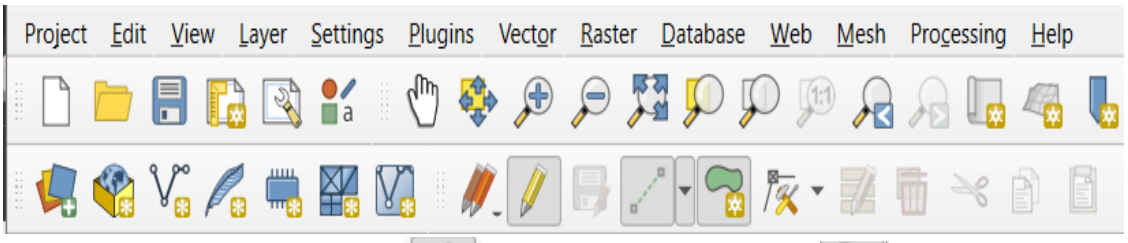


- The CRS dialog box will appear on screen. Click on the WGS*$ option and it will be selected as follows. Click on **OK**

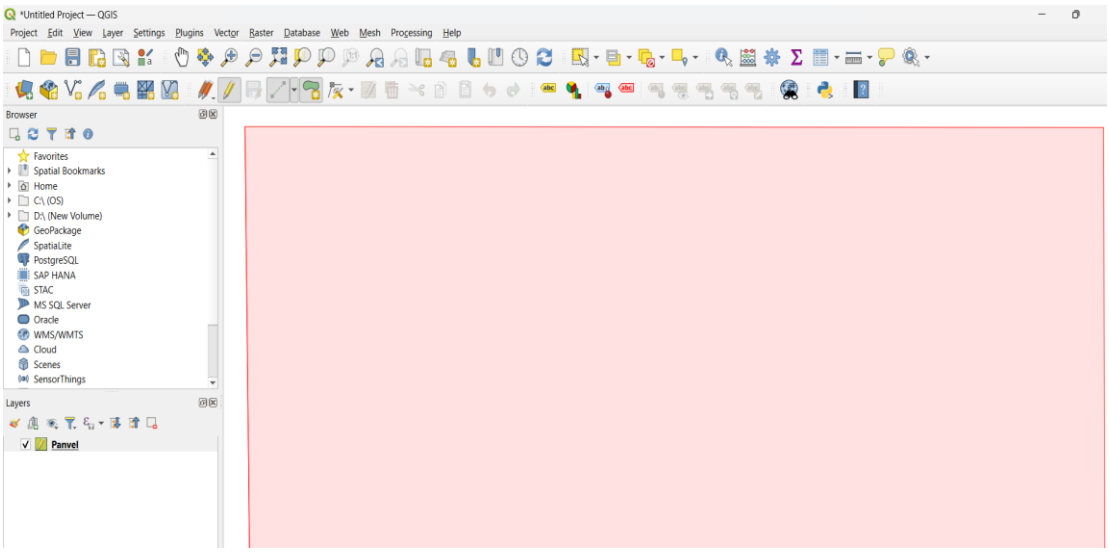


A. Follow the steps to plot **Polygon Features**.

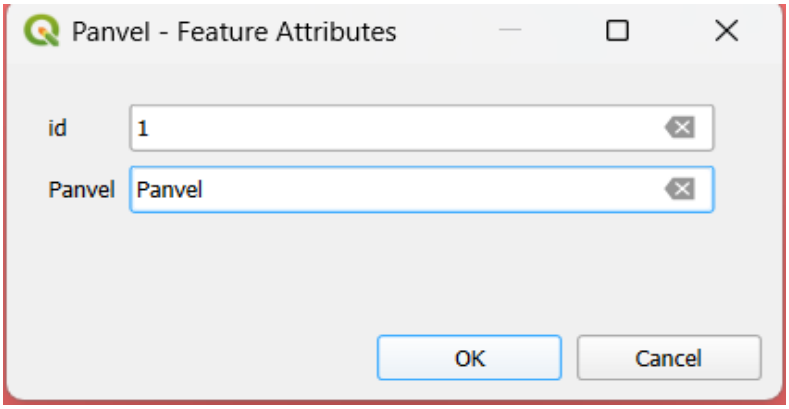
- Select the **Polygon Feature**(In our case it is **Matunga** for **Background**)from layer panel



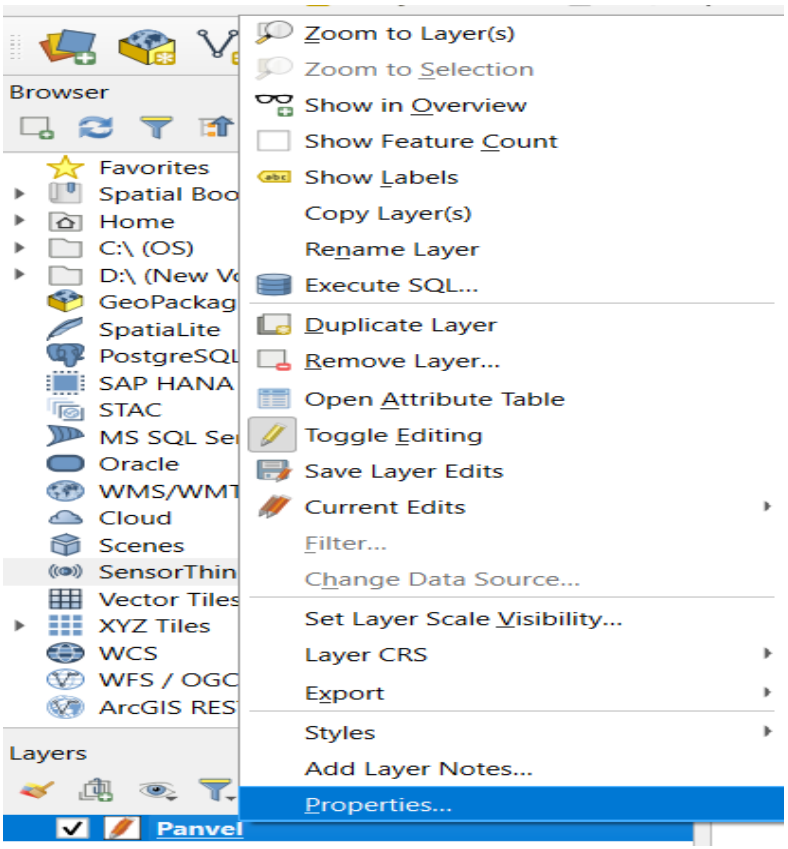
- Click **Toggle Editing Button** ->Click on **Add Polygon** ->Now place the cursor at the location where you want to place **the polygon layer minimum 3 points** should be selected.



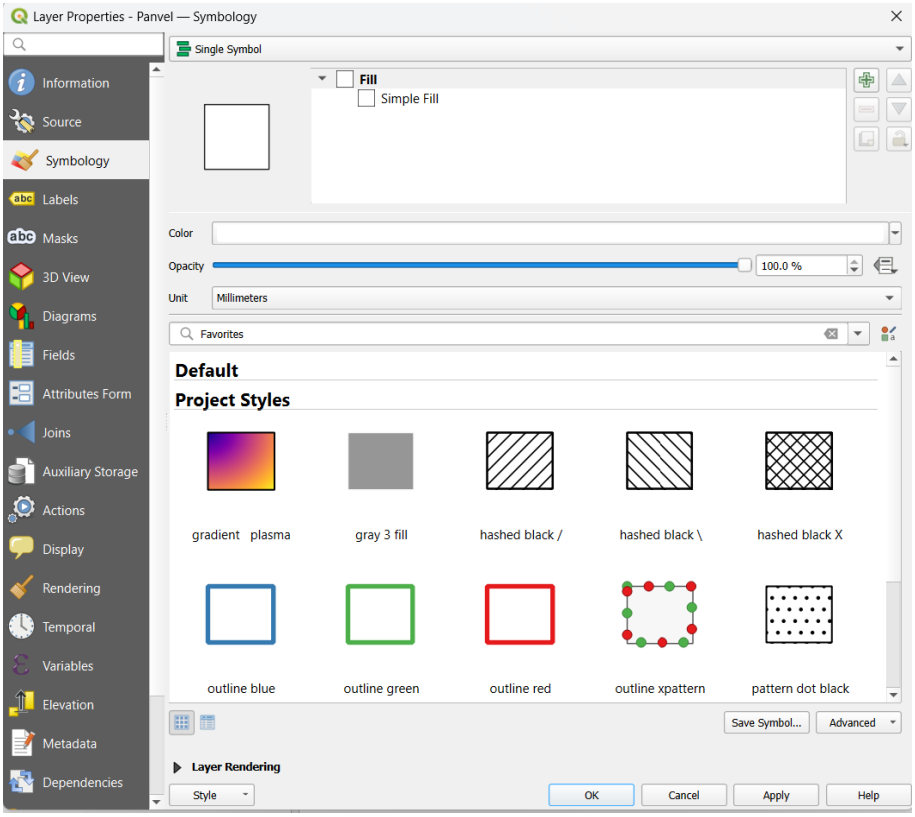
- Save the newly added polygon as follows.



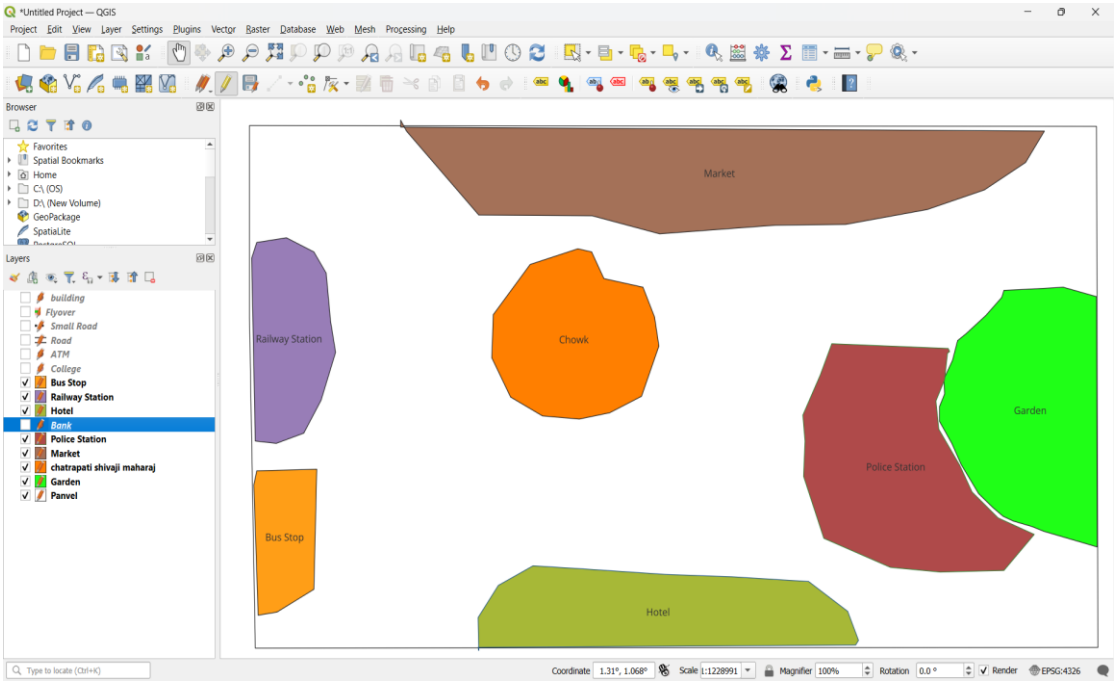
- Set **style** for polygon by using property window(**Right click** on Matunga Layer)



- Following screen will appear on the screen. Select **pattern** as you want and **click** on **OK.**

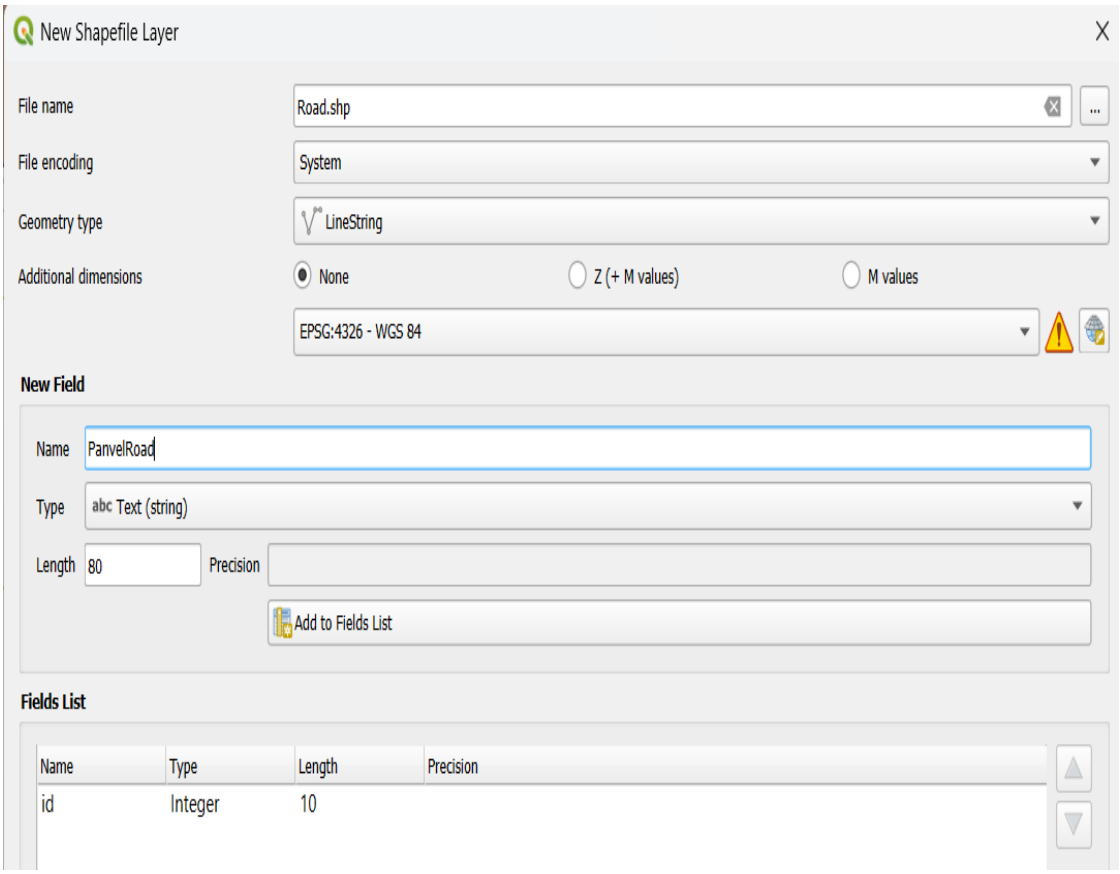


- Same way we can add one more polygon layer for Gardens.



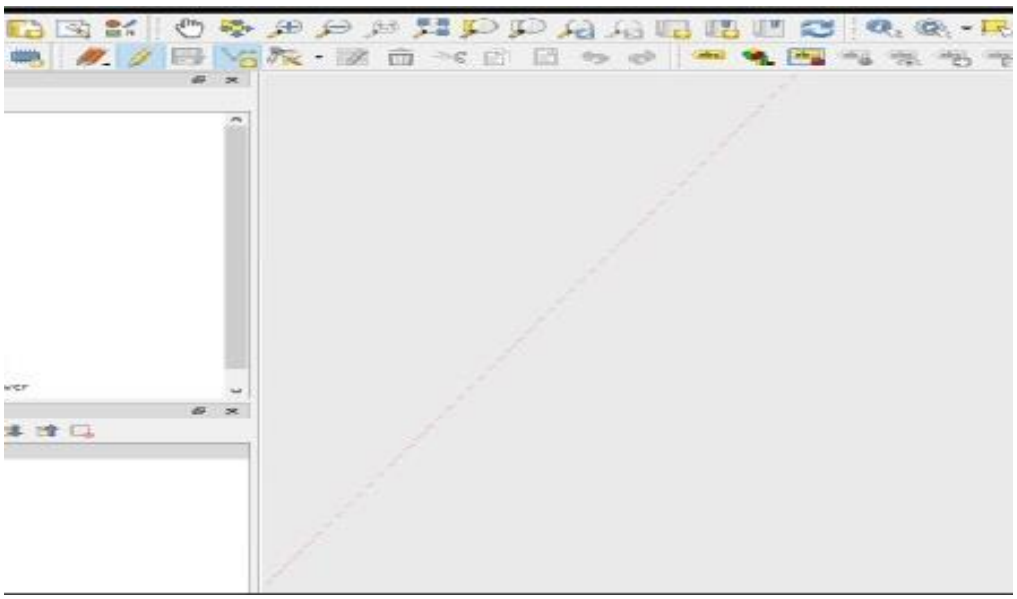
B. Creating Line Vector Layer

- Repeat the same steps as we have done for polygon layer.
- Select geometry type line.



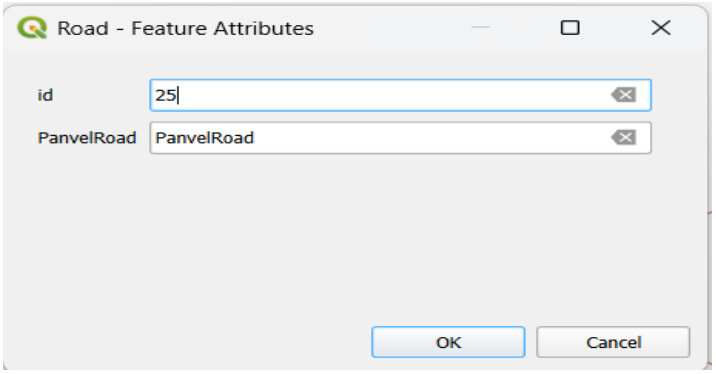
- Road Layer:
  - To plot road click on **Add Line Feature.**

  - Click on the map where you want to draw line.

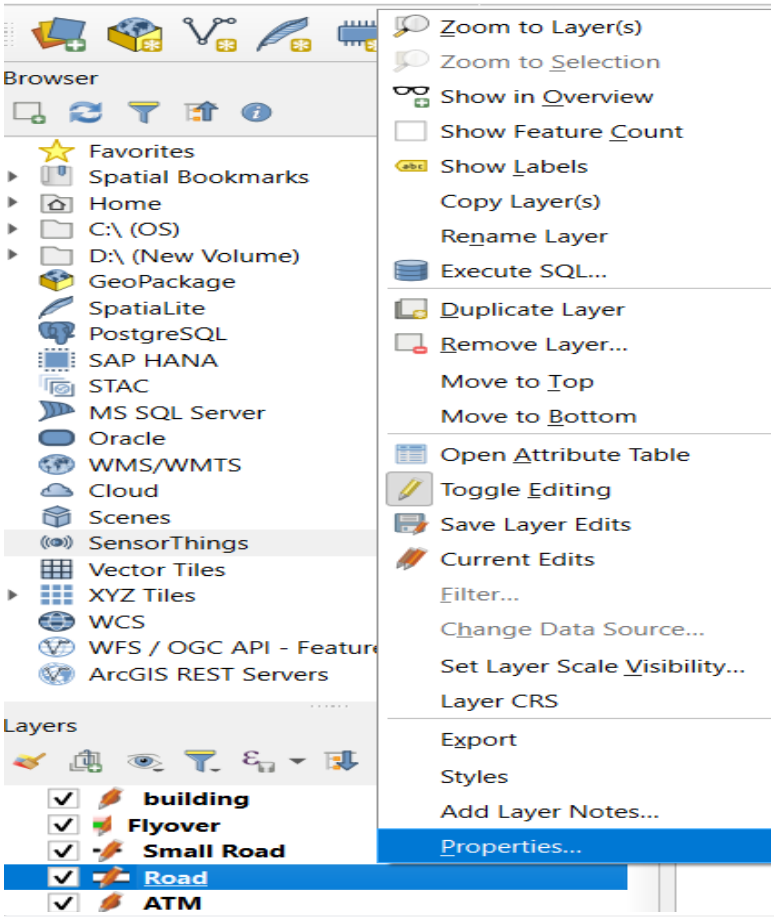


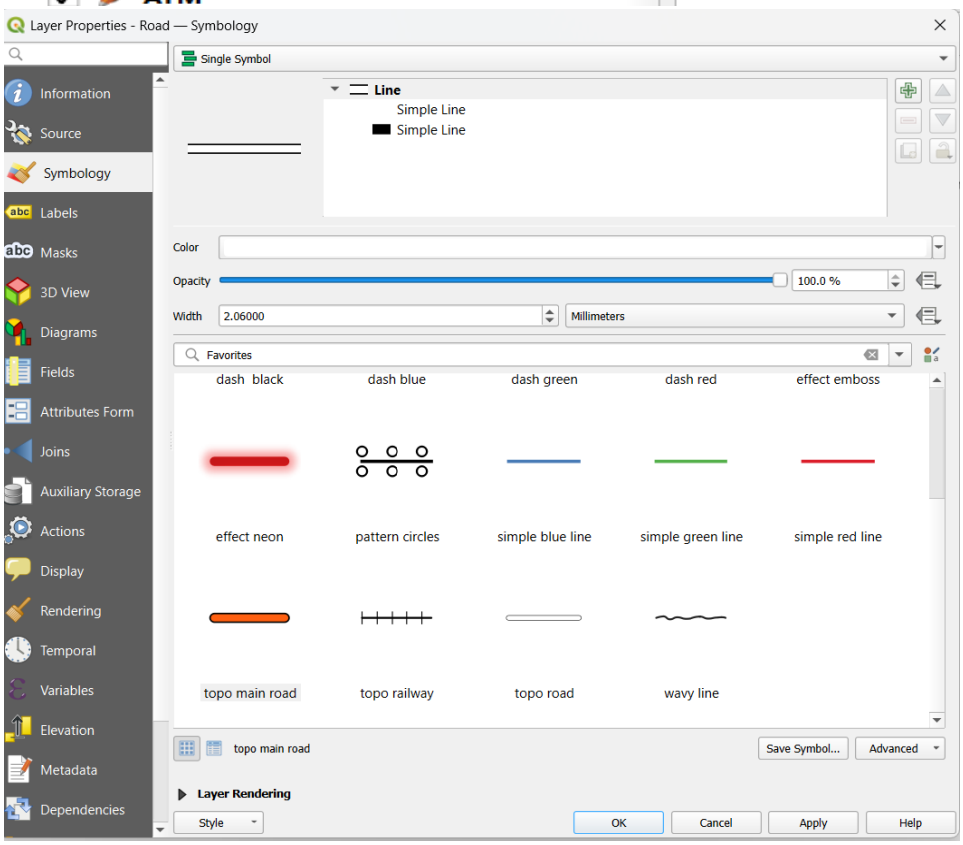
  - Once you are done then **Right Click on** map(**Dotted Line** turn into **Solid Line**)

  - **Save** your data

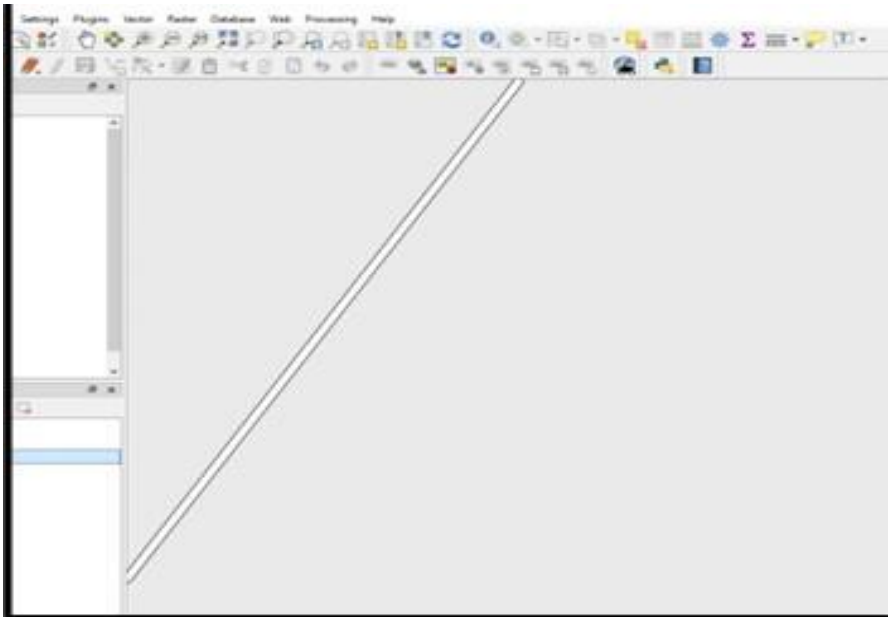


- Set **style** for roads in the same way as we have done for polygon.

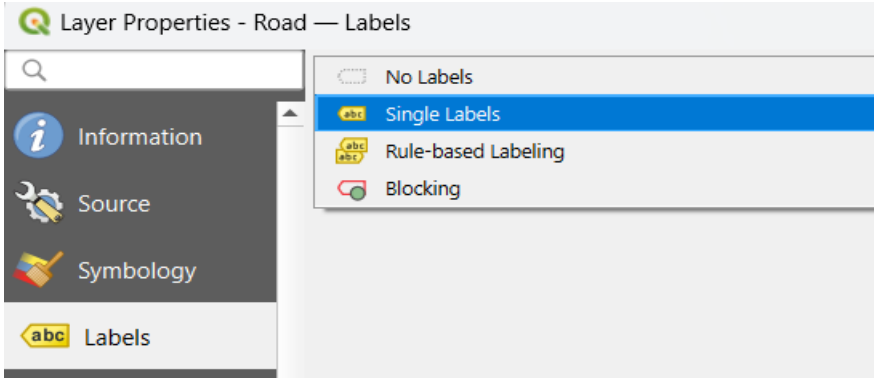




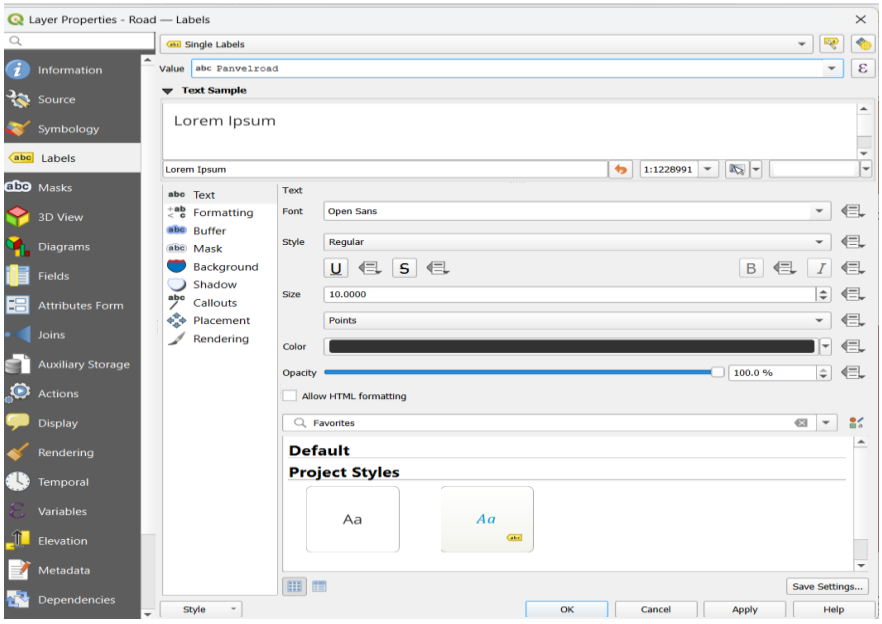
- Road will look as below.



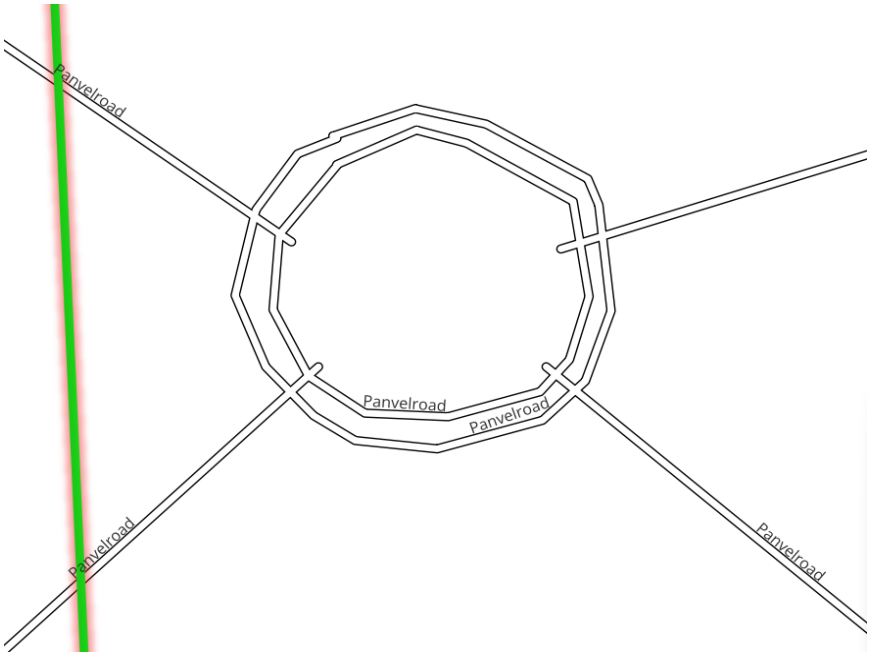
- To label your roads **Right Click** on **Road Layer**. Go to **Properties** window then select label and set single **Label Property.**



- Following window will appear on the screen

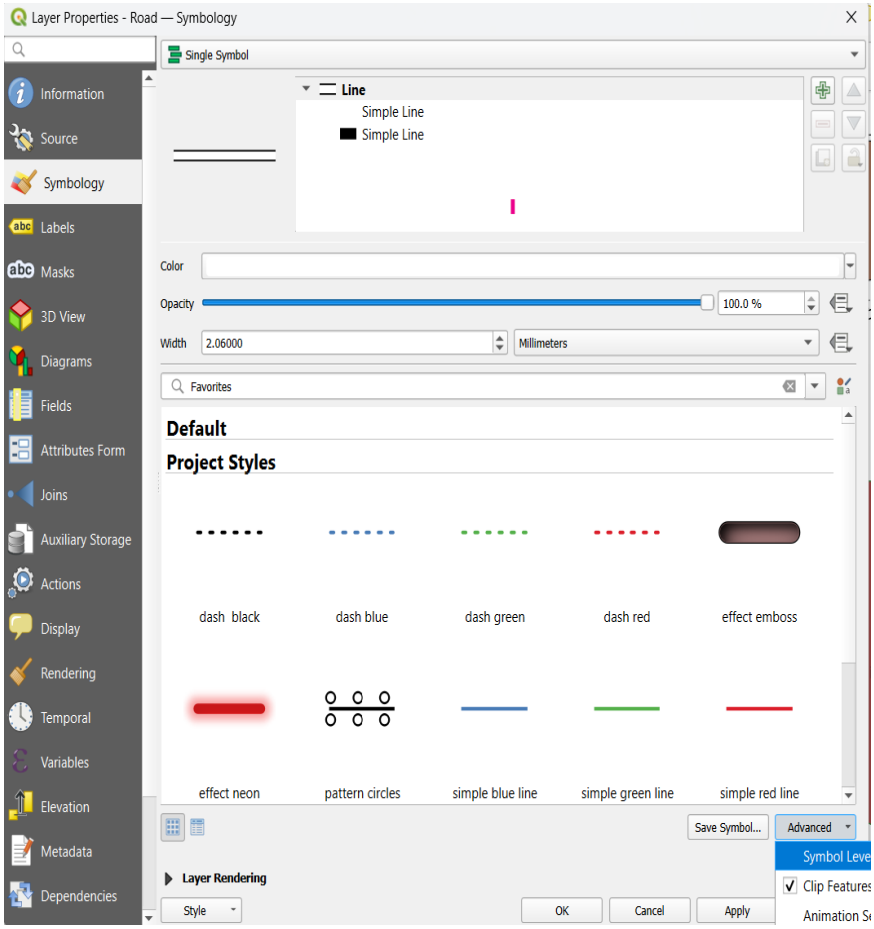


- Roads will look like these

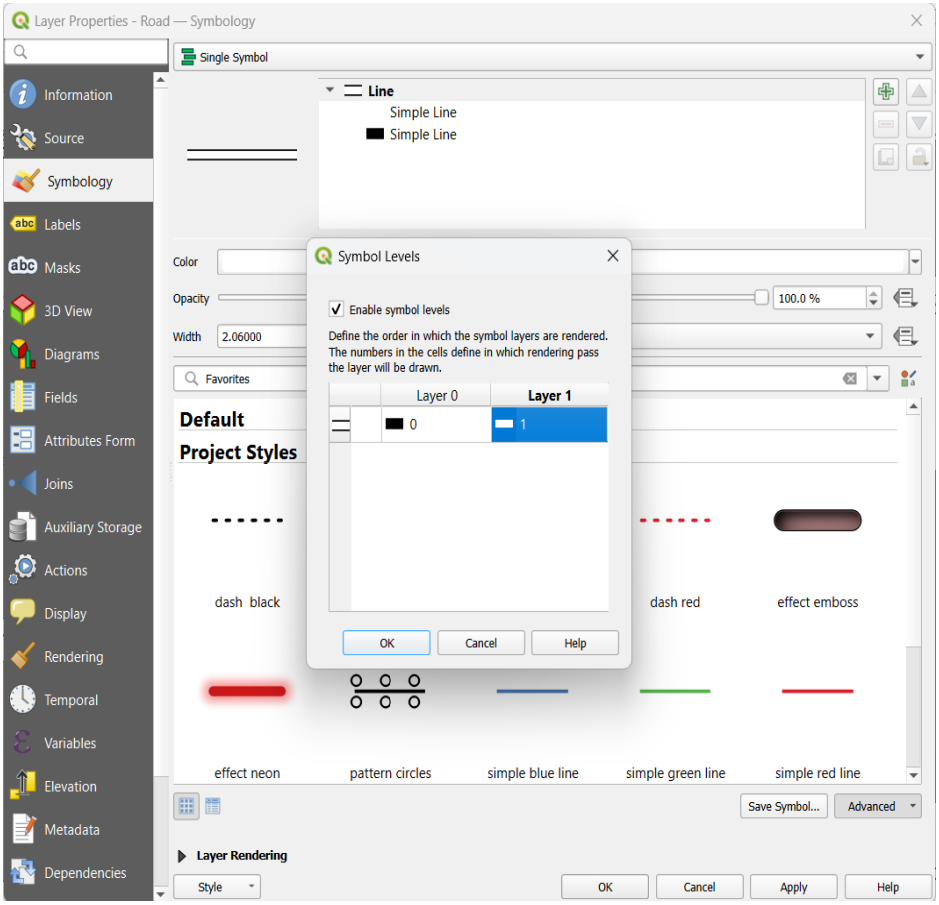


- To merge roads:

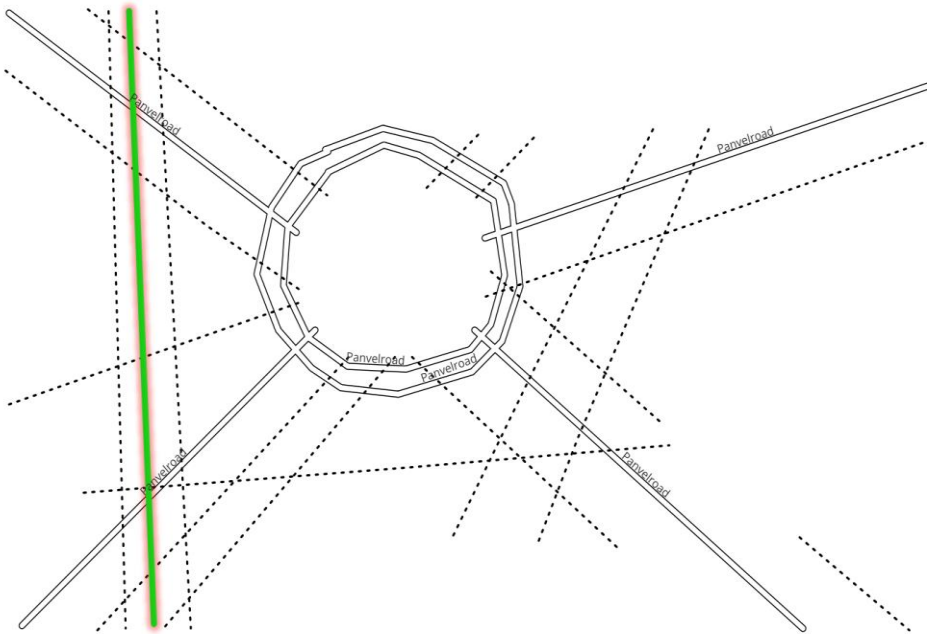
- Go to **Properties** of road then select **Symbology** . Click on **Advanced Button** select symbol levels.



- Check **Enable symbol levels** option.

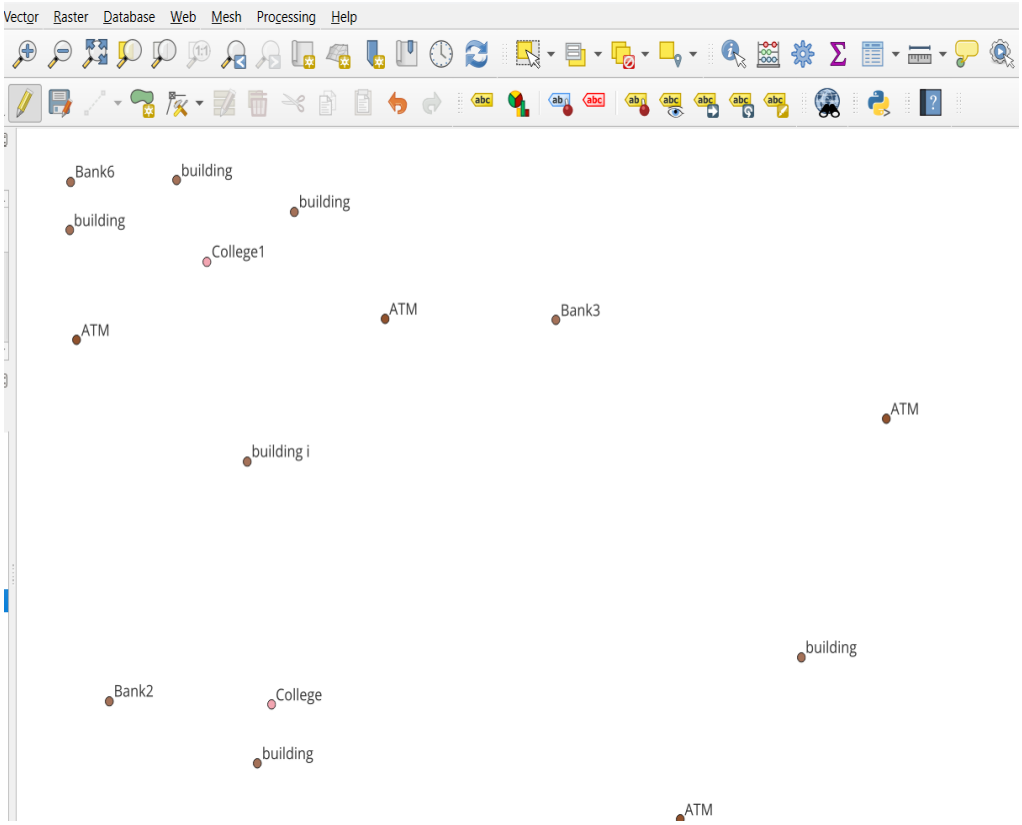
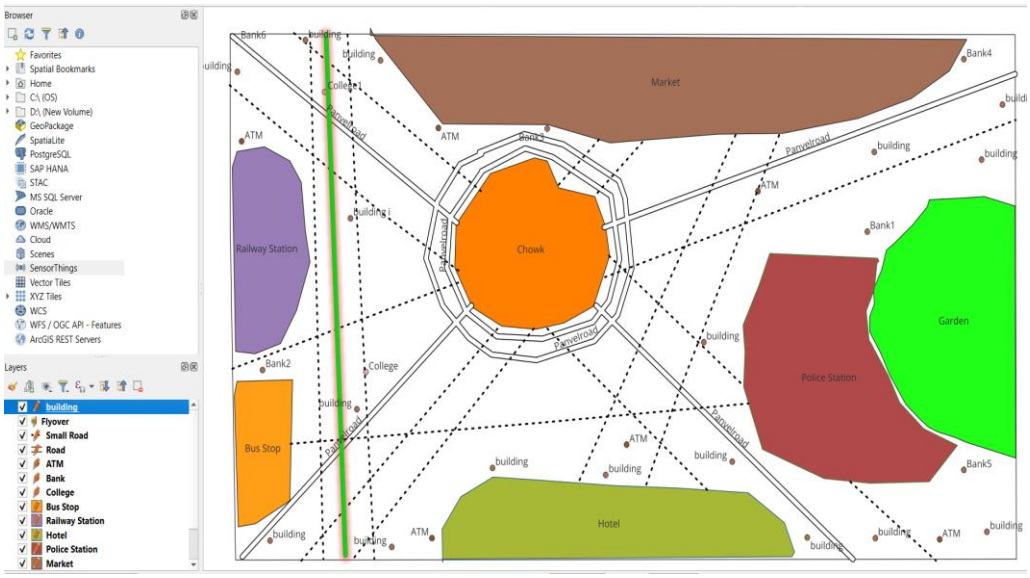


- Click **OK** & Road will appear as follows



C. Create Point Vector Layer

- Repeat same steps to add point layers as we have done in previous layers.(For ATM, Restaurants, Banks, Bus Stop etc.)



Final Output: